

An Investigation into the effect of thickeners and surfactants on the rate of creaming in an emulsion

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Link to repository: <https://github.com/br482-code/emulsion-tracker>

Abstract

Emulsions are thermodynamically unstable and only kinetically stabilised. This means that formulation is just about slowing down an inevitable break down. Stability testing is a bottleneck, and creaming index (CI) is a standard proxy.

This investigation tracked the CIs of 13 cylinders. These were all 20% oil-in-water emulsions, and they were tracked over 259 hours. A xanthan series (X) and a polysorbate series (E) were used. Rate constants were then extracted from the cylinders via an exponential-approach model.

It was found that the polysorbate reduced k monotonically from 0.0678h^{-1} to 0.0030h^{-1} . The xanthan was non-monotonic, in fact it first increased to 27% above the control, before decreasing to 0.0032h^{-1} .

It was also observed that the 0.20% xanthan and the 1-2% polysorbate converged on the same rate constant.

Creaming rate alone therefore cannot distinguish a thickener that slows droplet movement from a surfactant that reduces droplet size, despite the two acting on different destabilisation mechanisms.

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Introduction

The goal of this investigation was to analyse emulsions with different concentrations of thickener and surfactant added to them. The principle behind this was the key idea that emulsions are thermodynamically doomed but are kinetically rescuable – a stable emulsion is actually just one that decays slowly but using different chemicals we can lengthen out the timeframe of decay in order to give the appearance of a “stable” emulsion.

This is important industrially – shelf-life prediction is a genuine bottleneck in formulation. Customers will not buy products that look unappealing or like they might not work. For that reason, it is essential that emulsions hold together for a longer period of time so that is a larger window for them to be sold. However, perfect stability testing takes many months to years, as it requires continuous observation of the product over a long timeframe. This is not a practical option for companies, so the industry relies on accelerated testing and proxy measures. One such proxy is creaming index, which is what we are studying on a smaller scale in this investigation.

A formulator has two options to slow down the kinetic breakdown of the emulsion. The first option is to slow down the droplets by thickening the continuous phase – this is what our thickener (xanthan gum) achieves. The other option is to attack the cause by making the droplets smaller and harder to merge – this is what our surfactant (polysorbate 80) does.

Now that we have two mechanisms that both slow decay, we can proceed with the experiment in order to determine which lever buys more kinetic stability. We can also analyse the data to see whether the concentration-response of each behaves as Stokes’ law predicts.

To do this, we set up a series of 20% oil-in-water emulsions and record volumes at intervals. We can then use this to observe the change in creaming.

Theory

Destabilisation

The emulsion is affected by destabilisation. However, destabilisation is not one mechanism, but a collection of 4 competing ones which operate on different timescales. Thus, we need to understand which mechanisms actually occur in the experiment we conduct.

The four types of mechanism are as follows:

Creaming	Droplets migrate under gravity
Flocculation	Droplets clump together but don't merge
Coalescence	Droplets fuse as the film between them ruptures
Ostwald Ripening	Big droplets grow at the expense of small ones

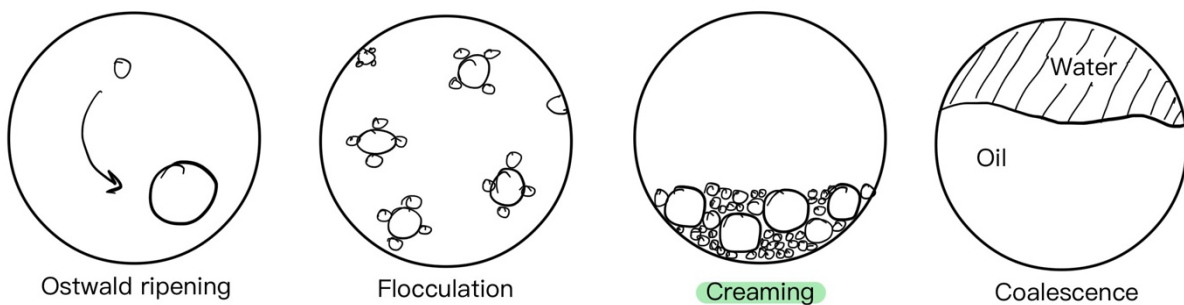


Figure 1 – The four types of destabilisation mechanism

The key difference between them that allows us to isolate creaming is that creaming (and flocculation to some extent) is reversible. The other mechanisms do not shift the serum boundary so the method used does not directly see them. This is what allows us to measure creaming, but also will include the effects of flocculation (as we see in certain results).

Stokes' Law

We can then analyse the results by comparing them to Stokes' law. Stokes law gives the terminal velocity of a sphere moving through a viscous fluid under gravity, reached when the buoyancy force is balanced by viscous drag, and can be seen in Figure 2.

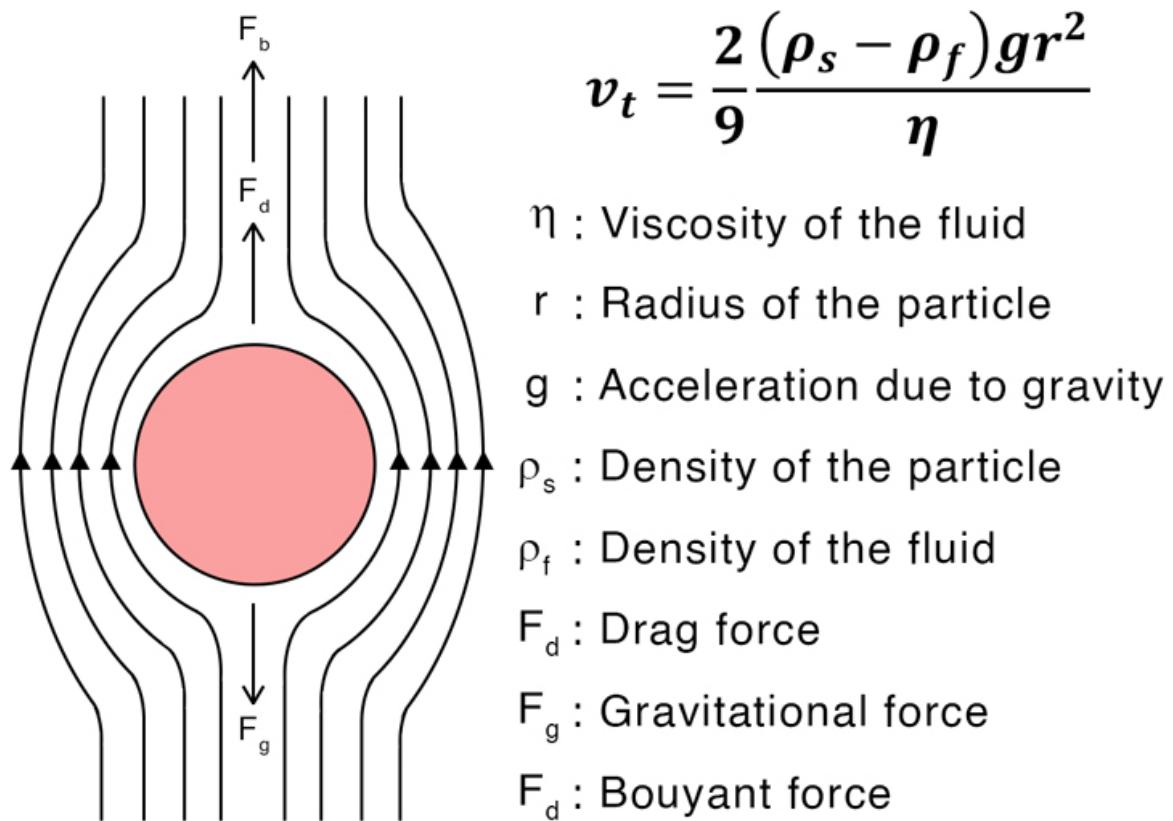


Figure 2 – Stokes' Law and what each variable means

One thing to note about Stokes' law is that it applies to isolated, rigid, spherical particles at a low Reynolds number. These are assumptions that we have applied to the system, but will be discussed in a later section as they are clear limitations of the investigation.

The main point to note about Stokes' law is that while r is squared, η is not, meaning that droplet size dominates viscosity as a lever. Changing r means acting at the interface, whereas changing η means acting on the bulk of the continuous phase. We can use this to compare the thickener and surfactant, as they are different routes.

The Kinetic Lever

Thickeners are the kinetic lever. A thickener is a substance that raises the viscosity of a liquid without substantially altering any of its other properties. The thickener used for this investigation was xanthan gum. Xanthan gum is used regularly in cosmetic products, so accurately simulates effects on real emulsions. Xanthan gum is a branched polysaccharide with structure as shown in Figure 3.



A thickener never touches the interface so it slows the approach of droplets but does not raise the barrier to their merging. This means although it buys time, it does not buy stability.

The other lever is the interfacial lever, caused by the addition of a surfactant. A surfactant (short for surface-active agent) is a molecule that adsorbs at an interface and lowers the interfacial tension there. The defining feature of a surfactant is structural – it's amphiphilic, with a hydrophilic head and a lipophilic tail (as seen in *Figure 4*). This results in it being energetically unfavourable to reside in the bulk of either phase so it must stay at the phase boundary.

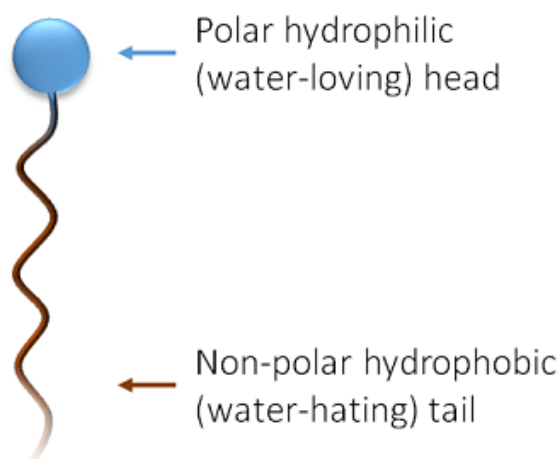


Figure 4 – the general structure of a surfactant

The surfactant chosen was Polysorbate 80. When the emulsion is blended, addition of the surfactant lowers the interfacial tension: this means that less energy is required to create new area so the same 30s of blending produces more surface area.

It then stabilises sterically because its polyoxyethylene chains are hydrated and project into the water^[1]. This means that when two droplets approach closely those chains are forced to interpenetrate, which is entropically unfavourable because the chains now lose configurational freedom. It is also osmotically unfavourable because the water floods back in to dilute the crowded region. Both results cause the droplets to be pushed apart.

The surfactant also has a self-repair mechanism called the Gibbs-Marangoni mechanism: as liquid drains between two touching droplets, it drags adsorbed surfactant sideways, leaving a patch of locally higher interfacial tension. Surfactant then flows back up that gradient, dragging liquid with it, re-thickening the film.

The Hydrophile-Lipophile Balance

Surfactants have a property called HLB – the Hydrophile–Lipophile Balance. This is a scale from 0-20, where a low number indicates a lipophile and a high number a hydrophile. The HLB is a scale devised to make emulsifier selection systematic rather than trial-and-error^[2]. The scale operates on different sections referring to different types of surfactants:

HLB	Application
3-6	Water-Oil emulsifier
7-9	Wetting Agent
8-16	Oil-Water emulsifier
13-15	Detergent
15-18	Solubiliser

Table 1 – the HLB scale

Polysorbate 80 has a HLB value of 15^[3] so is in the 8-16 band (as can be seen in *Table 1*). Thus, it is suitable for this investigation.

Oils have their own empirically determined HLB required value and in commercial industry formulators would match to it. However, for this investigation the oil used (sesame oil) has a required HLB value of 7^[3]. A more suitable surfactant would be a blend of polysorbate and sorbitan (colloquially called a Span/Tween blend) but this investigation uses solely polysorbate 80 as a single surfactant allows us to analyse just one variable without adding more complexity.

The main limitation of HLB is that it is solely computed from the structure of a molecule so does not consider other factors such as temperature, concentration and what type of oil is used.

What Stokes' Law predicts

From the equation, we can see that the only variables that change are r and η as we used one oil and assumed one constant temperature so the other letters are constants. It was mentioned earlier that droplet size dominates as r is squared but η is not. This suggests that the interfacial lever should be stronger, resulting in our samples with surfactant in them producing a lower rate of creaming. Stokes' Law predicts a constant velocity, so one might think that that the rate of creaming is linear. However, real creaming rate decreases as the cream layer crowds, so an exponential rate is preferred.

A predicted complication

One mechanism that can take place is called depletion flocculation. This is an effect where at a certain concentration the amount of non-adsorbing stabiliser can cause the droplets to push together. It is researched that at higher concentrations both additives can promote flocculation, so there is potential for depletion flocculation to occur^[4].

Methods

Design principle

We adjust one variable at a time from a shared control anchor. In this case the control was labelled C0, and concentration was altered between each sample while the other variables were kept constant. This is complicated by the main confounding factor – homogenisation. This is because it sets droplet size. To combat this, a control factor of using the same tool (the frother) on the same setting for 30 seconds was used. Using this identical method for all 14 cylinders helped prevent the confounding factor from significantly affecting the setup.

Materials and reagents

14 tall 100ml cylinders were used – due to supply issues two different brands of cylinders were used. However, this does not affect the investigation as creaming index is a ratio so self-normalises for diameter. The odd brands of apparatus were distributed across the different formulations as well in order to further level the disparities. A milk frother was used to homogenise the formulations, and 10ml pipettes were used for each different chemical. A single bottle of sesame oil was used, and the water source was tap water. The thickener used was xanthan-gum, and as this is a food-grade chemical it was easy to obtain in pure form. The surfactant used was Polysorbate 80. Polysorbate 80 was chosen due to being a defined compound with a documented HLB, unlike lecithin – another surfactant but one which has a HLB that is batch-dependent.

Stock solutions

First stock solutions were made for the thickener and the surfactant. For the xanthan stock, 1g of xanthan was measured on a scale with resolution of 0.1g. This was then added to 100ml of water. The xanthan powder had to be sprinkled in slowly in order to prevent it producing “fish eyes” as it hydrated. When left for an hour with only some intermittent stirring, the xanthan stock fully hydrated to form a viscous solution. The polysorbate stock is made by combining 10mL of polysorbate 80 with 90mL of water, and only required gentle stirring of the mixture, while producing no foam.

The formulations

The formulations were made according to the table shown below:

Label	Oil /mL	Xanthan stock /mL	Polysorbate stock /mL	Water /mL
C0	16	-	-	64
X1	16	4	-	60
X2	16	8	-	56
X3	16	16	-	48
E1	16	-	4	60
E2	16	-	8	56
E3	16	-	16	48

Table 2 – the formulations used in the experiment

The general rule was 20% oil and 80%, and additives were used to replace. This was done to produce 0.05/0.10/0.20% xanthan solution and 0.5/1.0/2.0% polysorbate solution. Each

formulation was made twice – an “a” and “b” sample. This was done to make the test fairer. One notable step was that each set of a and b samples were made separately, so that if there was a discrepancy in the batches it could be noticed.

Preparation and mixing

For each formulation, the aqueous reagents were all combined and then 16mL of oil was added. Then a milk frother was used to blend together all these reagents to produce an emulsion. The milk frother was used for an exact time of 30 seconds, to make all the emulsions equally as blended. Then the pour time was recorded and set at t=0 and the CI was defined as 0 at the start, as this is crucial to getting accurate data. The frother was then cleaned between samples to prevent any cross-contamination.

The cylinders were all kept in the same position for the whole of the experiment and were kept deliberately out of sunlight.

Creaming Index

Creaming index is a ratio and is written as a decimal. It is defined as the volume of serum divided by the volume of the total, as can be seen in *Figure 5*. As the cylinder has a uniform diameter throughout the whole graduated region, we can use height ratio as a replacement for volume ratio. Readings were taken at the creaming boundary – this is the interface between the lower translucent serum layer and the turbid emulsion above it. The serum volume was read directly off the graduations.

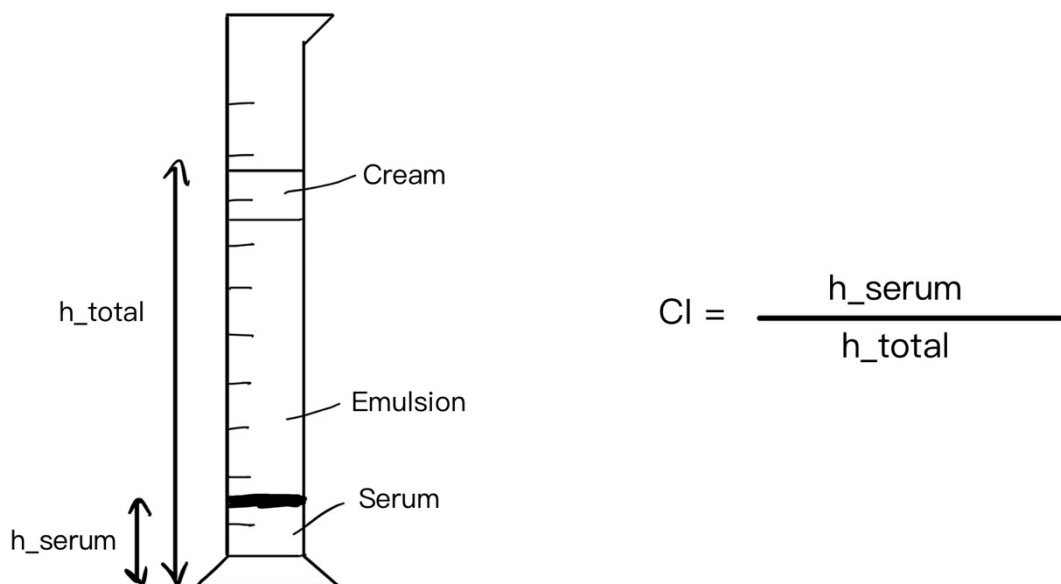


Figure 5 – the appearance of the cylinders when measured and the CI expression

Reading protocol

One issue lies in how the boundary appears – rather than being a sharp boundary there were two other possibilities for how the boundary could appear. The first was a diffuse boundary. To deal with this, the reading was taken at the midpoint of the diffuse boundary in all cases. The other case involved 3 sections being formed within the column – commonly seen in the surfactant series. For this, the bottom boundary was the one that was recorded, as this was the boundary between the serum and the emulsion. The readings were made at eye-level in order to avoid parallax error, and a constant backlight was used to produce the same lighting for each observation. All of the readings were made by a single observer which removes inter-observer error. The graduations are at divisions of 1mL, and all recordings were made to the nearest 0.5, producing an error of ± 0.006 CI units.

Recording protocol

The readings were stored in a spreadsheet with one row per cylinder per reading. The spreadsheet also stored the clock time and the elapsed hours since the cylinders' own pour. This is significant as the cylinders weren't poured simultaneously.

Rate model and fitting

The model used was:

$$CI(t) = CI_{\max} (1 - e^{-kt})$$

$CI_{\max}$ is the creaming index the emulsion approaches once separation is complete. k is the rate constant, and time is measured in hours. We can also use $1/k$ – the characteristic time – as this is the point at which CI has reached about 63%. This characteristic time can be used to calculate the half-life which is useful as a quotable figure. This model is used as Stokes' Law gives a constant velocity so predicts linear creaming, but in reality creaming decelerates as the cream layer crowds. The exponential form captures Stokes-like behaviour early and captures saturation late.

When fitting, a tempting form was to rearrange the equation for $-kt$ and to then fit that to a straight line. However, that requires assuming $CI_{\max}$ in advance and that would have distorted the error structure so this approach could not be used. Instead, both parameters were fit simultaneously and convergence was checked, using scipy.

Each cylinder was fit independently, giving two values of k per formulation (as averaging would have hidden the batch-to-batch variability that is the reason why the duplicates exist). The standard deviation was then computed so that it could inform us on how reliable our data was.

Results

Model Selection

The model used was $CI(t) = CI_{\max} (1 - e^{-kt})$ which has 2 parameters. However, we have to manually select a ceiling for this model – $CI_{\max}$ – as some of the graphs do not plateau. Thus we calculate a ceiling by thinking about packing. First, there is 16mL of oil and 64 mL of water, leading to a maximum of 0.8. However, applying the knowledge that random close packing occupies a volume of 0.64^[5], we finally get that $CI_{\max} = 0.69$. This is the geometric upper bound on what $CI_{\max}$ can be.

However, when the model was run on C0a with $CI_{\max} = 0.69$, k had a relative uncertainty of 19%. The model also systematically ran below the real data for the first 100 hours as can be seen in *Figure 6*. These were both observations that forced a change to be made to the model as it was not accurate enough.

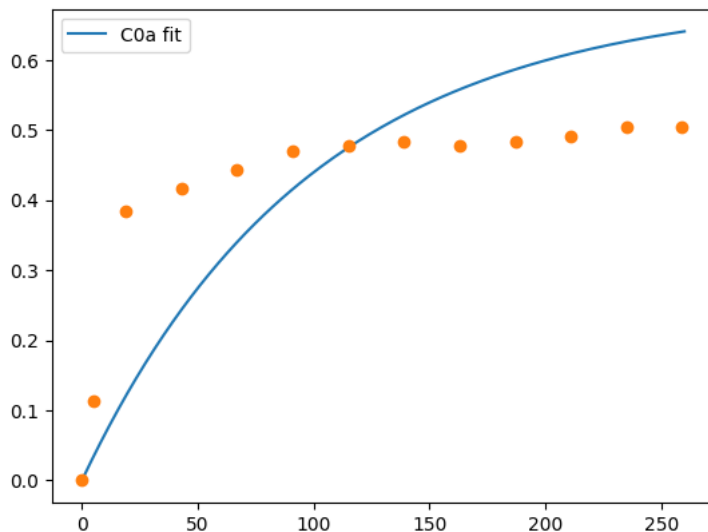


Figure 6 – the inaccurate first model

Thus a free-ceiling model was used – this is where $CI_{\max}$ is constrained within bounds of 0 and 0.69, but can vary within this range. The hard upper limit prevents the model from breaking on the formulations which do not plateau. The $CI_{\max}$ values for the plateaued cylinders all clustered within 0.44-0.52 – this shows that the amount of creaming was below our theoretical maximum, showing that although the geometry allows for a greater creaming layer this does not occur. This model is shown in *Figure 7* where it can be seen that it fits the data more accurately.

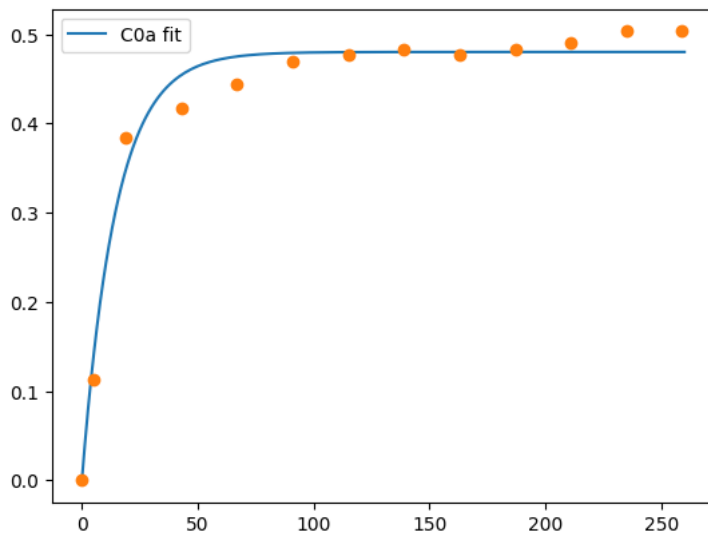


Figure 7 – the more accurate free-ceiling model used

Course of the experiment

The 14 cylinders were poured with staggered start times – each of which were recorded. By the first reading 5 hours later, the C and X series had already started to separate, and CI was already at 0.11-0.16 across the duplicates. In contrast, the E series had no clear creaming line to record as separation had not fully started yet.

Over the course of the investigation, cylinder X3a had to be excluded as it was disturbed due to human error. Its data was fully discarded so that there wouldn't be an external factor affecting a data series. This left behind only 13 cylinders and only 169 readings.

The E cylinders originally had layers with foam along them – this made the readings less accurate. E2a required use of the midpoint convention due to having too diffuse a boundary on a number of occasions.

The run was stopped at 259 hours into the investigation. This was due to microbial growth in the cylinders – this contaminant disrupted the layers in the cylinders and was also unsafe to allow to keep developing. The final readings taken show that the highest CI was 0.543 (from X1a).

Creaming Trajectories

All of the groups rose at first and then decelerated, as can be seen in *Figure 8*. However there are two general cases – one is the groups that rose and then plateaued, whereas the other is the groups that did not plateau before the end of the investigation at 259 hours.

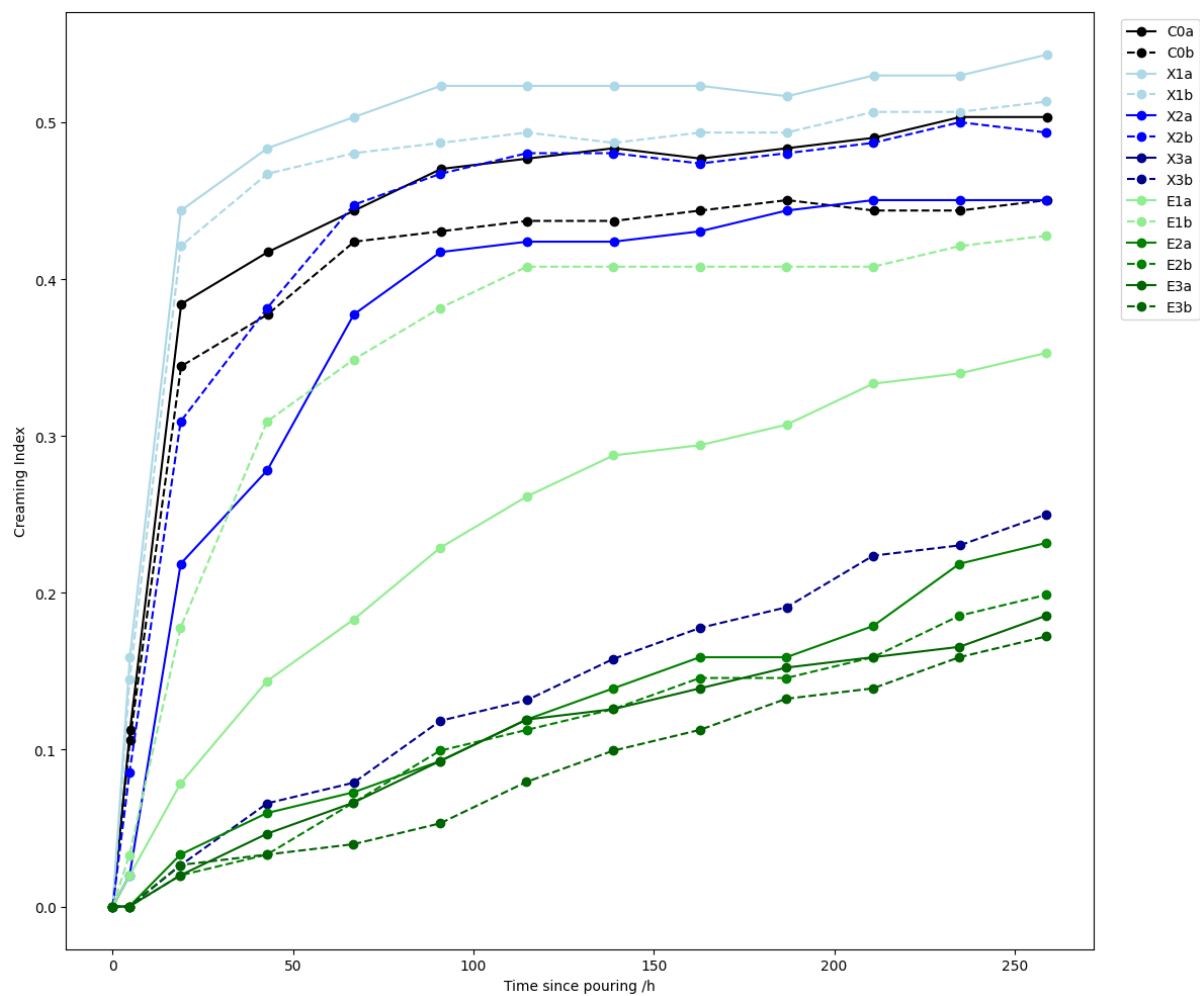


Figure 8 – all the replicates superimposed on one set of axes for comparison

C0, X1 and X2 rose rapidly over the first 20 hours but flattened by 100 hours. One interesting thing to note is that X1 rose to a higher creaming rate than the other 2 formulations so the xanthan series is non-monotonic in concentration.

X3, E2 and E3 climbed at an approximately linear rate and were still increasing at the cut-off time.

Finally, the duplicates usually mirrored each other well – however E1 had notable differences between the two cylinders with E1b reaching 0.42 by 100 hours whereas E1a was still rising at 200 hours.

Fitted Parameters

Fitted rate constants spanned 0.0864 to 0.0032 h^{-1} across the investigations which is a difference of a factor of 27. Formulation means with replicate spreads are given in *Table 3* and *Figure 9*.

Formulation	Conc.	k (h^{-1})	Spread	T _{1/2} (h)	Plateaued
C0	0	0.0678	0.0005	12.0	Yes
X1	0.05	0.0864	0.00001	10.4	Yes
X2	0.10	0.0363	0.0129	18.2	Yes
X3	0.20	0.00322	N/A	103	No
E1	0.50	0.0200	0.0130	44.1	Yes
E2	1.00	0.00296	0.00076	101	No
E3	2.00	0.00316	0.00293	107	No

Table 3 – the values of parameters from the investigation

NB no spread was found for X3 as X3a was discarded.

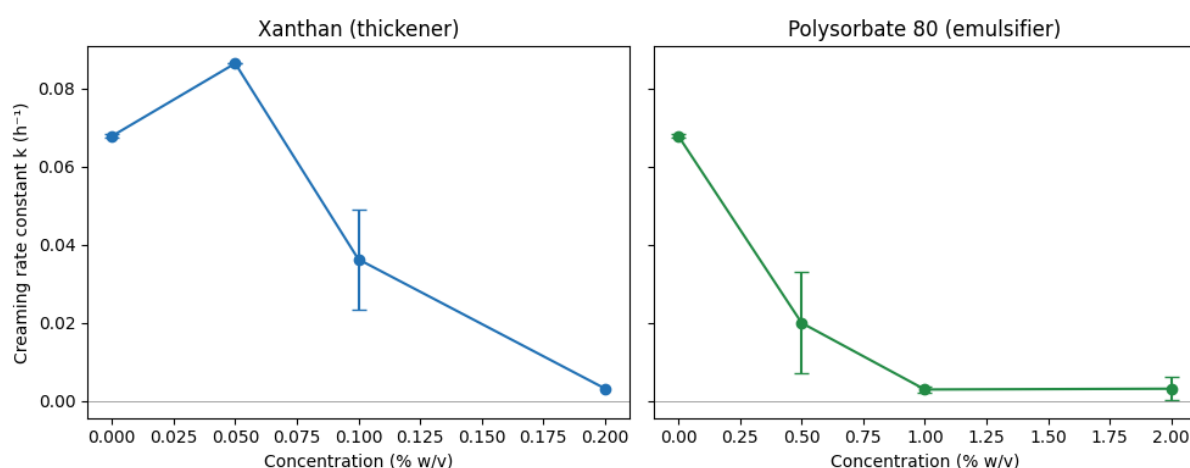


Figure 9 – a plot of k against concentration for both types of additive, with error bars

From this, we can see some clear results: X1 exceeded the C0 control (0.0864 vs 0.0678 h^{-1}); E2 and E3 fell within each other's spread (0.0030 and 0.0032 h^{-1}). Replicate spread was largest for X2 and E1. Finally, the fitted CI_{max} ranged from 0.24 to the 0.69 bound.

Fit quality and cross-check

As X3, E2 and E3 had not plateaued by 259 hours, their values for k and CI_{max} are correlated and individually poorly determined, as can be seen in *Figure 10*. Thus, whilst we can use these values to rank the different formulations, the individual k values should not be relied upon.

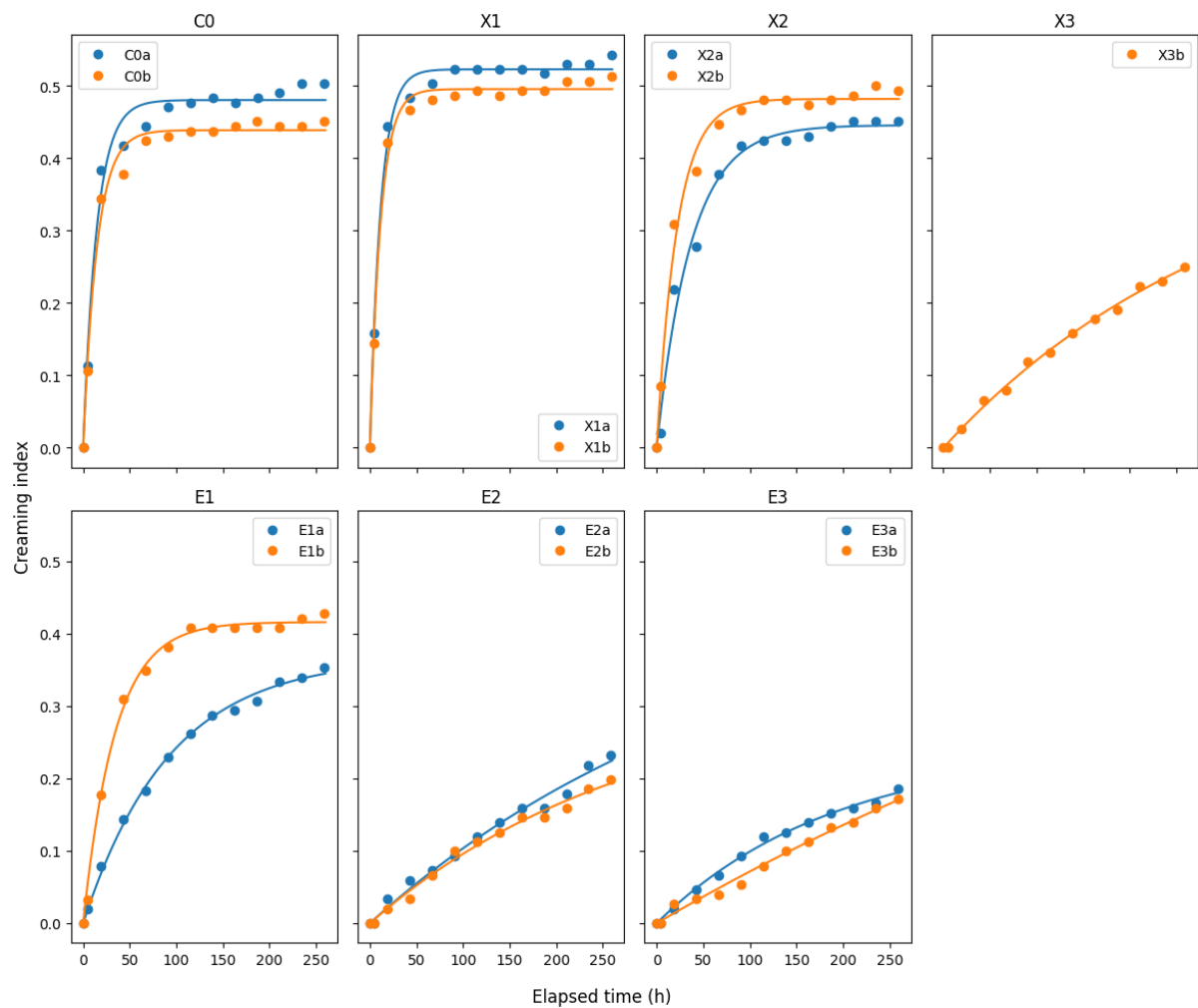


Figure 10 – 7 plots, each with just the series in question plotted to show the disparity between replicates. These also show that k and Cl_{max} are correlated for the non-plateaued formulations.

The model is supported by the cross-check conducted using $t_{1/2}$. The model-free $t_{1/2}$ ranks the cylinders in almost the same order, with the only exception being the swapping of X2a and E1b. However, this still shows that even with our adapted model with the free value for Cl_{max} the results still are mostly accurate.

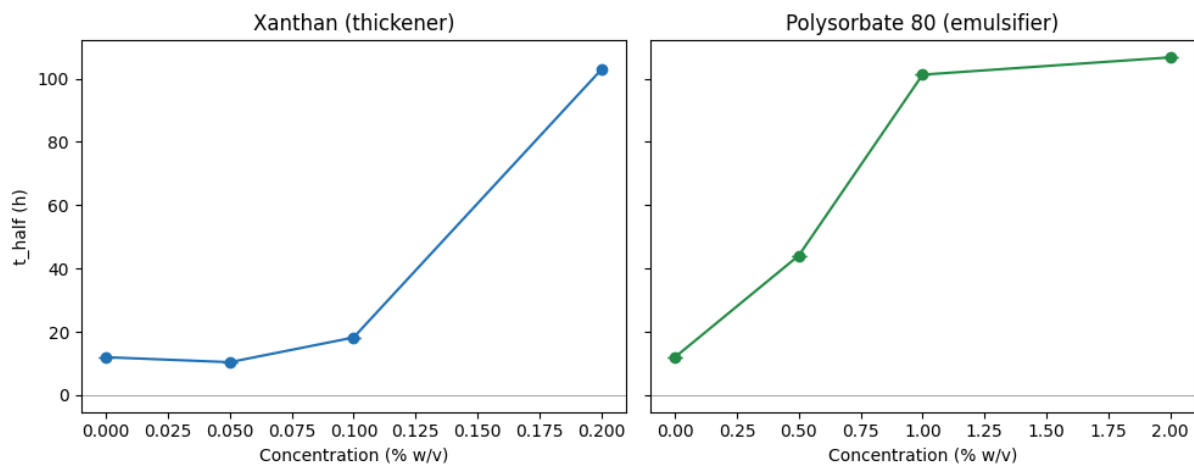


Figure 11 – the $t_{1/2}$ model, showing that the results are similar in both cases.

Discussion

The Xanthan Reversal

Theory flagged that depletion flocculation was a mechanism that could cause either series to behave in a way that doesn't match our predictions, and the xanthan series shows this. While the control had a $k = 0.0678\text{h}^{-1}$, after 0.05% xanthan was added, k was raised to 0.0864h^{-1} . This shows that the rate of creaming was faster when this much xanthan was added, before it decreased for the 0.1% xanthan case. Both the X1a and X1b batches agree to 3 significant figures on the value of k and agree to 1% error on $t_{1/2}$ suggesting that these observations are real rather than products of error.

The observations appear consistent with the idea that depletion flocculation occurs as a mechanism. It occurs because the xanthan coils don't adsorb so there is a depletion layer around each droplet from which the coils are geometrically excluded^[4]. This leads to layers overlapping when droplets approach each other, causing the coils to gain accessible volume and hence entropy, causing free energy to drop. Flocs rise as single large objects, and due to the r^2 term in Stokes' Law, a floc creams far faster than its constituent droplets.

However, this is only significant for the X1 formulation – this is because xanthan is raising η at the same time. By X3 (0.20% xanthan) viscosity dominates and k collapses as depletion flocculation is no longer the more significant effect.

The Polysorbate series

We observe that in the E series k falls monotonically and steeply – but E2 and E3 are indistinguishable at 0.0030 and 0.0032 h^{-1} .

One might assume that the reason for change in the E series is due to changes to the interfacial tension, however we know that this is not true as E1's CMC (critical micelle concentration) is several 100 times smaller than its concentration.

In reality coverage capacity affects the series. Blending produces large amounts of interfacial area, and so there must be sufficient surfactant to cover the new area created. This results in smaller droplets, so a smaller r in Stokes' Law.

One noticeable issue is the replicate problem – E1a and E1b differ by a factor of 3. This is consistent with the 0.5% concentration of being near the coverage threshold, where small differences in the air caused by shear in certain batches produce different droplet populations.

Where the Levers meet

Both 0.20% xanthan and 1-2% polysorbate 80 arrive at the same rate constant despite using different mechanism and being added in differing concentrations. This equivalence is misleading as they arrive there by acting on different terms in Stokes' Law. Xanthan raised η

whereas polysorbate reduced r . This means the two formulations are only equivalent in their final creaming rate, not in how they achieve this goal.

The practical consequences of choosing a different additive differ. Using a thickener gets a product that separates at the same rate but is thick, and whose stability would be expected to be destroyed when the continuous phase is sheared or diluted – these interactions however do not affect a surfactant-stabilised interface. However, we cannot distinguish which additive has been used by calculating creaming rate, as this does not distinguish them.

Limitations

This investigation has a few limitations, especially due to the fact it was conducted in an environment that was not a laboratory, so some equipment that could have been used to make the experiment more accurate was not available.

What the method does not measure

One significant limitation is that droplet size is never measured – r is inferred from the outcome rather than directly observed. A slower rate constant is consistent with the mechanism proposed but doesn't establish it.

Coalescence is also not directly observed, so inference has been made from a reversible proxy. In addition, the effects of depletion flocculation cannot be confirmed or excluded without microscopy or a redispersion test, so an anomalous rate constant would have more than one possible explanation.

Another limitation is specific to X3 – due to human error X3a was disrupted. This invalidated its results, meaning we have no replicates available for the X3 test so cannot compare and verify the results obtained from X3b.

Measurement precision

The weighing scale had a resolution of 0.1g, meaning when measuring 1g there was a $\pm 10\%$ error propagating identically in all three samples. This means there is a systematic offset in the concentrations of the formulations. In addition, the manual reading of the measuring cylinders comes with error due to having to read to 0.5mL scale, and because a manual judgment must be made on where the boundaries are, and how to deal with diffuse boundaries too.

Design

The experimental setup only had duplicates of each formulation, rather than three copies. This means that a standard deviation is less reliable than if there had been 3 replicates. Formulations that have rate constants that differ by less than the replicate spread cannot be separated because of this.

Some other absences from the design of the investigation include the fact that there was no temperature control. Due to the lack of a laboratory, there was a variable temperature that was not recorded. However, η is temperature-sensitive, so ambient drift is uncontrolled.

Scope

There are some limitations in the scope of the investigation conducted. Only using one type of oil, one type of surfactant and one type of thickener means we cannot compare the effects of multiple species of additives. In addition, we are using ambient conditions of only testing for 11 days- this is because no preservative was included so after less than 2 weeks microbial growth developed. However, an ideal shelf life would be a year or more and stress testing under accelerated conditions.

Using a milk frother to mix the formulations is also not ideal, as the property required of good apparatus for this investigation would be for it to have a higher shear and lower aeration than a

frother – as in my investigations the droplet sizes will be much bigger than ones run through a commercial homogenizer.

Finally, there are some clear limitations relating to the assumptions made about Stokes' Law. The droplets are crowded and fluid, as opposed to our assumptions that they are isolated and rigid. This means quantitative agreement with Stokes' Law is not expected and only the ranking of rate constants can be tested.

Further Work

There are higher level techniques that would be ideal in the investigation in order to counter the limitations previously identified.

One example of this is droplet sizing. In the investigation, r is inferred from the outcome. Being able to photograph a diluted sample on a slide at $t=0$ would be incredibly useful. This would allow us to create a droplet-size distribution and compare the predicted Stokes' velocity against the fitted k .

Another example would be using a redispersion test to distinguish flocculation from coalescence. Inverting a cylinder at the end of the test would then allow us to see whether the original CI trajectory was reproduced. This would then show proof of flocculation, as flocs disperse under agitation, whereas coalesced droplets do not.

Conclusion

Over the course of 259 hours, the creaming indexes of 13 cylinders were tracked continuously. The rate constants were then derived from an exponential approach-fit model.

It was discovered that for the polysorbate series (E) k fell monotonically and saturated above 1%, consistent with the interface reaching coverage.

For the xanthan series (X) k was not monotonic – 0.05% xanthan raised k above the additive free control (C) before higher concentrations caused it to decrease. This result is consistent with depletion flocculation competing with the increase in viscosity.

Finally, it was observed that 0.20% xanthan and 1-2% polysorbate both converged on the same rate constant, showing that even if two different mechanisms are used the creaming rate can still be decreased by the same amount. This indicates that creaming rate cannot be used to distinguish the thickener that slows droplet movement from the surfactant that reduces droplet size.

References

- [1] Sigma-Aldrich. TWEEN® 80, Product Number P 5188 — Product Information Sheet.
<https://www.sigmaaldrich.com/deepweb/assets/sigmaaldrich/product/documents/393/290/p5188pis.pdf>
- [2] Griffin, W. C. "Classification of Surface-Active Agents by 'HLB'." *Journal of the Society of Cosmetic Chemists*, vol. 1, 1949, pp. 311–326.
<https://library.scconline.org/v001n05/8>
- [3] HLB Calculator. List of Oils and Emulsifiers with HLB Values.
<https://hlbcalc.com/materials>
- [4] Grundy, M. M.-L., et al. "Influence of oat components on lipid digestion using an in vitro model: Impact of viscosity and depletion flocculation mechanism." *Food Hydrocolloids*, 2018. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6020131/>
- [5] Ni, R., Stuart, M. A. C., Dijkstra, M., and Bolhuis, P. G. "Pushing the glass transition towards random close packing using self-propelled hard spheres." *Nature Communications*, vol. 4, 2013. <https://www.nature.com/articles/ncomms3704>